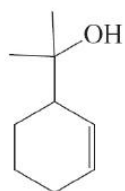
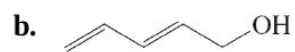
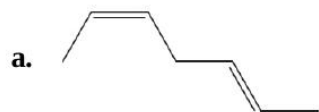


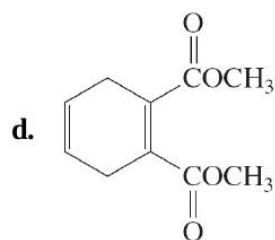
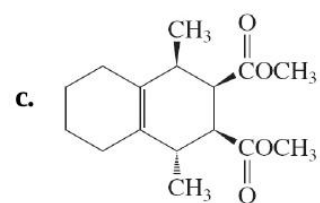
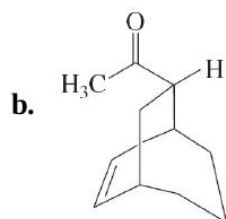
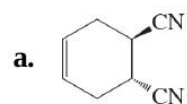
44. Starting with cyclohexene, propose a reasonable synthesis of the cyclohexene derivative shown below.



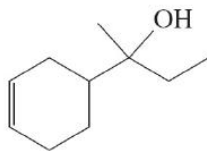
45. Give a systematic name to each of the following molecules.



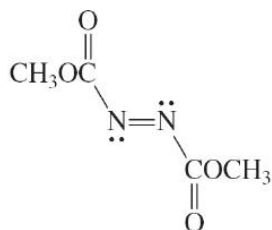
58. Propose a synthesis of each of the following molecules by Diels-Alder reactions.



61. **CHALLENGE** Propose an efficient synthesis of the cyclohexenol below, beginning exclusively with acyclic starting materials and employing sound retrosynthetic analysis strategy. [**Hint:** A Diels-Alder reaction may be useful, but take note of structural features in dienes and dienophiles that permit Diels-Alder reactions to work well ([Section 14-8](#)).]



62. Dimethyl azodicarboxylate (see below) takes part in the Diels-Alder reaction as a dienophile. Write the structure of the product of cycloaddition of this molecule with each of the following dienes. **(a)** 1,3-Butadiene; **(b)** *trans, trans*-2,4-hexadiene; **(c)** 5,5-dimethoxycyclopentadiene; **(d)** 1,2-dimethylidenecyclohexane. Ignore the stereochemistry at nitrogen in the products (amines experience rapid inversion, as we shall see in [Section 21-2](#)).



Dimethyl azodicarboxylate